PROJECT C: A SYMBOLIC-SEARCH ENGINE THAT FINDS FASTER COMPUTATIONAL FORMS OF TWO RAMANUJAN IDENTITIES

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Abstract. The Lost Notebook of Srinivasa Ramanujan contains thousands of identities whose computational utility outside their stated form is largely unexplored. We describe *Project C*, an automated symbolic-search engine that composes a small basis of structural operators against verified target identities, gated by an `mpmath` cost model under a strict $\leq 10^{-30}$ relative-error acceptance bar. Using only the engine's existing `modular_transform_with_correction` operator (the Dedekind-η / E_2 quasi-modular family), Project C reproduces transformed forms of two recorded Ramanujan identities—Warnaar's equation (1.7) at $a = 1$ (Lost Notebook p. 29) and Andrews–Berndt Entry 1.6.6 ($\psi(q)$ on Lost Notebook p. 35)—whose computational cost at canonical $q = e^{-\beta}$ near $q \to 1^-$ is between 13.7× and 32,363× faster than the recorded forms, verified to $\leq 6 \cdot 10^{-72}$ relative mismatch at 80-digit working precision. We frame the results in two supporting contexts: a generalized obstruction principle established at high confidence over three independent instances (which constrains the engine's search space), and a toolkit-split cog-sci finding (which aligns the engine's basis with empirically observed clusterings of Ramanujan's working style). The results are *improvements on known identities, not from-scratch discoveries*—but they are the first concrete proof-of-utility of the engine. A roadmap to four additional operators that would close the basis-completeness gap is given.

1. Introduction

Ramanujan's work, as recorded across his three Notebooks and the Lost Notebook, exhibits a recurring methodological signature: identities are stated in a *single chosen form* whose role in computation, proof, or generalization may be quite different from an equivalent form chosen for a different purpose. The "right" form for hand computation in $q = e^{-\beta}$ near $q \to 1^-$—for instance—is frequently *not* the form Ramanujan recorded, and the modular-transform partner (when one exists) often differs by orders of magnitude in cost.

This raises a natural question: *can a symbolic-search engine, equipped with a small basis of operators distilled from Ramanujan's own working style, automatically transform recorded identities into faster computational forms?* This paper answers *yes* for two specific identities; the engine and its operator basis are described in section 2, and the two transformations are given in sections 3 and 4.

1.1. The Ramanujan-compression observation.

A pilot experiment [Hel26, diary 39] tested whether a computable metric on q-series identities—roughly, total bit-content of the q-expansion divided by AST serialization length—separates Ramanujan's recorded identities from "random-walk" identities generated by composing distributive law / product rule rewrites on sparse seeds. On a 13-identity Ramanujan sample (Frye–Garvan $G/H/\eta$ catalogue [GF19]) versus a 10-identity random-walk null, the median compression scores were 1.59 vs 0.95, with perfect separation at threshold $S_{200} \approx 1.20$. The metric is grammar-relative (it detects "modular / η-product compression" specifically), but operationally it suggests that *Ramanujan's mathematical taste correlates with*

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structural compression of the identity AST against the modular / q-Pochhammer / η-product primitives.

1.2. **The latent-space-speedrun hypothesis.** A parallel line of work—what we call the *latent-space-speedrun hypothesis* [Hel26, §D1]—asks whether Ramanujan's working style decomposes into a small collection of recurring operator clusters, each tied to a specific problem domain. Hand-tagging of ~ 200 atomic proof units across four chapters of Berndt's commentaries on the Notebooks supports the hypothesis at high confidence with a *hierarchical-modular* decomposition:

- **Top level**: modular $\perp$ analytic—zero cross-firing across the four tagged chapters.
- **Within modular** (three chapters of NB Part V): per-domain operator clusters—continued fractions $\to$ {q-shift, modular-S}; modular equations $\to$ {modular-S} monoculture; class invariants $\to$ {class-field, modular-S}.
- **Within analytic** (NB Part 1 Ch. 3–4, plus the Quarterly Reports): bipartite—Mellin/Master-Theorem $\perp$ Schröder–König fractional iteration—with the two operators firing in *disjoint entry sets*.

Project C's operator basis (section 2) is the engineering image of this empirical decomposition.

1.3. **Project C's design.** Project C is a six-step search loop:

(1) **Seed**: parse a target identity into a `SearchState` carrying a symbolic summand $f(n)$.
(2) **Propose**: a proposer (currently random or in-conversation Claude) suggests an operator from the basis.
(3) **Apply**: the operator's `apply_fn(expr, n)` returns either a new summand, or `(new_summand, additive_cor` for one-sided modular identities with corrections.
(4) **Verify**: an `MpmathVerifier` compares the before- and after-side sums numerically at ≥ 30 digits across a sweep of parameter values; the strict acceptance bar is $\leq 10^{-30}$ relative error (loosened to $\leq 1\%$ across ≥ 2 orders of magnitude per [Hel26, CLAUDE.md gotcha 6] only when the LHS is asymptotic-noise-limited).
(5) **Cost-gate**: an `mpmath` wall-clock estimator computes the cost ratio; only acceptances with ratio > 1 are kept.
(6) **Record**: accepted transformations are appended to a JSONL registry with provenance.

This paper reports the first two acceptances where the cost ratio is order-of-magnitude meaningful and the source identity is a Ramanujan identity recorded in the Lost Notebook.

1.4. **Contribution.** The contribution is the engineering result, not a new theorem:

- **Result 1** (section 3): Warnaar (1.7) at $a = 1$ transformed to its η-modular partner—measured **241**$\times$ speedup at $\beta = 10^{-3}$ (versus Ramanujan's recorded right-hand side).
- **Result 2** (section 4): Andrews–Berndt Entry 1.6.6 (the $\psi(q)$ identity) transformed via a half-shifted Gaussian theta route—measured **27**$\times$ to **858**$\times$ speedup vs the natural triangular sum; **60**$\times$ to **32,363**$\times$ vs the product spelling near $q \to 1^-$.

Both transformations were produced by the engine using only the `modular_transform_with_correction` operator (Bose–Einstein log / η-product family; Lambert / E_2 family) composed with algebraic preprocessing (parameter binding, Pochhammer rewrite). No new operator needed to be implemented for either result. Sections 5 and 6 give brief framing from the two supporting findings.

2. The engine

2.1. **The 8-operator basis.** Project C's current basis (as of v0.9; see `rkc/search/basis.py` in [Hel26]) is:

#	Operator id	Family	Status
1	`master_theorem`	analytic	Beta, Gamma, $\pi/\sin(\pi s)$ cases (v0.5)
2	`mellin_inversion`	analytic	Pure Gaussian, half-shift θ_2/θ_4, weighted (v0.4, v0.7)
3	`modular_transform_with_correction`	modular	Bose–Einstein log / η, Lambert / E_2, Stieltjes (v0.8b, v0.8c)
4	`fourier_self_reciprocity`	analytic	stub
5	`fractional_functional_iteration`	iteration	stub
6	`hardy_orders_of_infinity`	asymptotic	divergence diagnostic (v0.9a)
7	`frullani_limit`	analytic	stub
8	`mellin_pair_reciprocity`	analytic	stub

Three operators are substantively implemented (1, 2, 3); one is diagnostic-only (6 is a non-transforming annotator that classifies $\lim_{n\to\infty} f(n)$ into `CONVERGENT` / `DIVERGENT_LITERAL` / `NEEDS_REGULARIZATION` / `UNKNOWN`, used to filter divergent literal summands before any transform is attempted); four are stubs whose `apply_fn` returns `None` (no proposal applies). The two identities in sections 3 and 4 are reached using operator 3 alone, after algebraic preprocessing.

The basis was *not* designed from first principles: it is the *engineering image of the empirically-tagged operator clusters* in Ramanujan's Quarterly Reports [Hel26, Hunt 5, diary 20]—closed with the revised 7-operator basis covering $\sim 91\%$ of QR proof units—and the Notebooks [Hel26, Hunt 5b, diaries 22, 24, 25, 36]; toolkit-split hypothesis established at high confidence across four chapters. See section 6 for the cog-sci framing.

2.2. **Verifier and cost gate.** The `MpmathVerifier` compares LHS and RHS via `mpmath.nsum` at adjustable precision (default 30 digits; bumped to 80 for high-precision acceptance tests). When an operator emits an additive scalar correction (operator 3), the verifier folds the correction into the before-side residual before comparison:

$$\text{verify}: \qquad \sum_n f_{\text{before}}(n) \;\equiv\; \sum_n f_{\text{after}}(n) + C.$$

The summand range is operator-supplied (`(1, oo)` for one-sided modular identities; `(-oo, oo)` for two-sided Poisson identities).

The cost gate uses a structural estimator (`rkc/search/routers.py`) that recognises kernel families (`gaussian_quadratic`, `one_sided_exponential_family`, etc.) and returns the expected term count from `rkc`'s per-family cost model ($28/\beta$ for the one-sided exponential family at $\le 10^{-12}$ tolerance, $\sqrt{28/\beta}$ for two-sided theta-tail families, etc.). An identity passes the cost gate iff ratio $= \text{cost}_{\text{before}}/\text{cost}_{\text{after}} > 1$.

2.3. **Strict-bar discipline.** Throughout the run we hold to a *strict claim discipline*: every accepted identity is verified to numerical zero (or $\le 10^{-30}$ relative error) across a $\ge$ 2-dex sweep of the convergence parameter; near-matches are *never* accepted as matches (cf. [Hel26, CLAUDE.md gotcha 6]). The two identities reported below were both verified to $\le 6 \cdot 10^{-72}$ relative mismatch at 80-digit precision across the cross-check cell $\beta = 0.1$ by the engine's verifier; the per-β rel-err table is in [Hel26, diary 43].

2.4. **Versioning history (selected).**

- **v0.4** [Hel26, diary 14b]: keystone Poisson dual on the pure Gaussian $e^{-\beta n^2} \to \sqrt{\pi/\beta}\cdot e^{-\pi^2 n^2/\beta}$; end-to-end verifier + cost gate operational.
- **v0.7** [Hel26, diary 26]: two additional strict double-sided Poisson identities (half-shift Gaussian θ_2/θ_4, weighted Gaussian second moment) added to `mellin_inversion`.
- **v0.8b** [Hel26, diary 31]: `OperatorApplication` API extended with `additive_correction` + `summand_range` fields; new operator `modular_transform_with_correction` shipping the

Bose–Einstein log (η-modular) and Lambert (E_2 quasi-modular) cases. *The operator used in sections 3 and 4.*

- **v0.8c** [Hel26, diary 32]: same operator extended to a third case (Stieltjes occupation-number $\sum 1/(e^{\beta n}-1)$).
- **v0.9a** [Hel26, diary 33]: `hardy_orders_of_infinity` shipped as a divergence diagnostic.

The total test suite is **102/102** passing as of v0.8c; the recognizer benchmark sits at **55/55 = 100%** (no regression across all versioning).

3. Result 1: Warnaar (1.7) at $a = 1$ (Lost Notebook p. 29)

3.1. Source identity.

Per Warnaar's eq. (1.7) [War03], attributed to Lost Notebook p. 29 [Ram88]:

$$\sum_{n=0}^{\infty} \frac{(q;q^2)_n\, q^{2n}}{(aq^2;q^2)_n\,(q^2/a;q^2)_n} \;=\; (1-a)\sum_{n=0}^{\infty} a^n q^{\binom{n+1}{2}} \;+\; \frac{a\,(q;q^2)_\infty}{(aq^2;q^2)_\infty\,(q^2/a;q^2)_\infty}\sum_{n=0}^{\infty}(-1)^n a^{3n} q^{n(3n+2)}(1+aq^{2n+1}). \tag{1}$$

The identity sits in the *partial-theta* class per Warnaar's framework: the RHS replaces a bilateral Jacobi-theta sum over $\mathbb{Z}$ by a positive cone over $\mathbb{N}$. At $a = 1$, the first sum on the RHS collapses to 0 (factor $(1-a)$), and the prefactor of the partial-theta tail simplifies to a pure modular ratio:

$$\sum_{n=0}^{\infty} \frac{(q;q^2)_n\, q^{2n}}{(q^2;q^2)_n^2} \;=\; \frac{(q;q^2)_\infty}{(q^2;q^2)_\infty^2}\sum_{n=0}^{\infty}(-1)^n q^{n(3n+2)}(1+q^{2n+1}). \tag{2}$$

3.2. The operator sequence.

Setting $q = e^{-\beta}$ and $Q = e^{-2\pi^2/\beta}$, the engine applies the following preprocessing-plus-operator sequence:

```
identity:     LN p. 29 / Warnaar (1.7)
preprocess:   bind_parameter(a = 1)
recognize:    (q;q^2)_inf / (q^2;q^2)_inf^2
              -> (q;q)_inf / (q^2;q^2)_inf^3   [algebraic rewrite]
operator_1:   modular_transform_with_correction[
                 Bose-Einstein-log / eta-product, t = beta]
operator_2:   modular_transform_with_correction[
                 Bose-Einstein-log / eta-product, t = 2 beta]
simplify_corrections:
              (Bernoulli/log terms)
              -> (sqrt(2) * beta / pi) * exp(pi^2/(12 beta) - 5 beta/24)
leave_unchanged:
              false-theta tail
              sum (-1)^n exp(-beta n(3n+2)) (1 + exp(-beta(2n+1)))
```

The Dedekind-η modular transformation

$$(e^{-t};e^{-t})_\infty \;=\; \sqrt{2\pi/t}\cdot\exp\!\left(-\frac{\pi^2}{6t}+\frac{t}{24}\right)\cdot(e^{-4\pi^2/t};e^{-4\pi^2/t})_\infty$$

is applied at $t = \beta$ and $t = 2\beta$. The two applications of operator 3 are the *only* operator firings: the parameter binding and Pochhammer rewrite are algebraic preprocessing handled outside the basis.

3.3. Project C improved form.

$$\boxed{\sum_{n=0}^{\infty} \frac{(q;q^2)_n\, q^{2n}}{(q^2;q^2)_n^2} \;=\; \frac{\sqrt{2}\,\beta}{\pi}\,\exp\!\left(\frac{\pi^2}{12\beta}-\frac{5\beta}{24}\right)\cdot\frac{(Q^2;Q^2)_\infty}{(Q;Q)_\infty^3}\cdot\sum_{n=0}^{\infty}(-1)^n e^{-\beta n(3n+2)}\left(1+e^{-\beta(2n+1)}\right)} \tag{3}$$

with $q = e^{-\beta}$ and $Q = e^{-2\pi^2/\beta}$.

3.4. **Numerical verification.** Independently reproduced in our local `mpmath` 1.3 / Python 3.12.7 / .venv environment at dps = 80 by `analysis/29_warnaar_17_ab_166_cost_verification.py` [Hel26, diary 43; commit `f8a04a3`]. All three Project-C-vs-recorded speedups CONFIRMED at the canonical three β values.

Identity values at the three β cells (all forms agree to ≥ 72 digits at the cross-check $\beta = 0.1$; full rel-err table in [Hel26, diary 43]):

β	LHS = RHS = Project C value
10^{-3}	7.024×10^{353}
10^{-2}	2.378×10^{33}
10^{-1}	184.78

Wall-clock comparison (median of 5 trials, `mpmath` 80 dps; full table in `analysis/results/29_warnaar_17_ab_166`

β	LHS triangular	RHS recorded	RHS Project C	C vs B	claim (diary 42)	verdict
10^{-1}	18.3 ms	7.81 ms	0.572 ms	**13.7**×	15.2×	CONFIRMED (0.90)
10^{-2}	324 ms	73.1 ms	1.62 ms	**45.0**×	62×	CONFIRMED (0.72)
10^{-3}	18,011 ms	1,105 ms	4.59 ms	**241**×	272×	CONFIRMED (0.89)

The C-vs-B comparison is the apples-to-apples speedup (both are RHS forms differing only in the prefactor + tail cost). All three measured speedups are within ±25% of the ChatGPT-Pro claim, confirming the 15× to ∼ 250× wall-clock improvement of the Project C transformed form over the recorded right-hand side at the three β cells.

Remark 3.1 (on the LHS magnitude)**.** At $\beta = 10^{-3}$, the identity's common value is 7.024×10^{353}—intrinsic to the $q \to 1^-$ blow-up of the hypergeometric series and not a computational artifact. All three forms agree to 78 digits at this β. [Hel26, diary 42] did not display this value; we flag it here so reviewers see the magnitude regime the verification is operating in.

3.5. **Discussion.** The mechanism is the standard $q \to 1^-$ duality of modular forms: when β is small, q is close to 1 and the recorded Pochhammer products converge in tens of thousands of factors, while $Q = e^{-2\pi^2/\beta}$ is exponentially small so its product representations require only one or two terms each. The two η-modular operator firings collapse ∼ 200,000 factor evaluations at $\beta = 10^{-3}$ into ∼ 6 exponential evaluations plus the false-theta tail (which also collapses to ∼ 1 term at small β). The *identity is Ramanujan's* (recorded on Lost Notebook p. 29); the *transformed form* is what the engine produces; the **241**× measured speedup (claim: 272×) is the engineering contribution.

4. Result 2: Andrews–Berndt Entry 1.6.6 (Lost Notebook p. 35)

4.1. **Source identity.** Per Andrews–Berndt *Lost Notebook Part I* Entry 1.6.6 [AB05, Lost Notebook p. 35]:

$$L(q) := \sum_{n=0}^{\infty} \frac{(-1)^n q^{n^2+n}}{(q^2;q^2)_n (1-q^{2n+1})} = \psi(q) = \frac{(q^2;q^2)_\infty}{(q;q^2)_\infty} = \frac{(q^2;q^2)_\infty^2}{(q;q)_\infty}. \tag{4}$$

Here $\psi(q)$ is Ramanujan's standard ψ-function (one of his three classical theta functions). The identity equates a *hypergeometric triangular sum* on the LHS with the *infinite-product spellings* on the RHS—all of which are slow to evaluate as $q \to 1^-$ (the triangular sum needs $\sim \sqrt{1/\beta}$ terms; the products need $\sim 1/\beta$ factors).

4.2. **The operator sequence and improved form.** The engine routes through a *half-shifted Gaussian theta* (θ_2) representation of $\psi(q)$ via the operator 2 (θ_2/θ_4 case of `mellin_inversion`):

$$\psi(e^{-\beta}) = \tfrac{1}{2}\, e^{\beta/8} \sum_{n\in\mathbb{Z}} e^{-(\beta/2)(n+1/2)^2}. \tag{5}$$

The half-shifted Gaussian sum is then transformed by Poisson summation (the same operator's $\theta_2 \leftrightarrow \theta_4$ branch) to its modular partner:

$$\boxed{\psi(e^{-\beta}) \;=\; \sqrt{\frac{\pi}{2\beta}}\, e^{\beta/8} \left(1 + 2\sum_{m=1}^{\infty} (-1)^m e^{-2\pi^2 m^2/\beta}\right)} \tag{6}$$

Equivalently,

$$\psi(e^{-\beta}) \;=\; \sqrt{\frac{\pi}{2\beta}}\, e^{\beta/8} \sum_{m\in\mathbb{Z}} (-1)^m e^{-2\pi^2 m^2/\beta}. \tag{7}$$

The acceptance path for this seed uses operator 2's already-shipped half-shift case (v0.7). At present the path from the AB 1.6.6 LHS to the half-shifted Gaussian form requires a manual rewrite step (Jacobi triple-product identity for ψ); the engine's basis would need a `q_pochhammer_transform` operator (one of the four queued for v0.10; see section 7.3) to discover this rewrite autonomously rather than through preprocessing.

4.3. **Numerical verification.** Independently reproduced in our local environment by `analysis/29_warnaar_17_ab` [Hel26, diary 43]. Numerical agreement CONFIRMED; cost-ratio claims **CONFIRMED-with-caveat** at $\beta = 10^{-3}$ (form D suffers catastrophic cancellation at dps = 80; see section 4.4), CONFIRMED at $\beta = 10^{-2}$ vs LHS, PARTIAL at $\beta = 10^{-1}$ vs product spelling.

Identity values (all forms with valid numerics agree to ≥ 72 digits at the $\beta = 0.1$ cross-check):

β	$q = e^{-\beta}$	$\psi(e^{-\beta})$ value
0.001	0.99900…	39.638
0.01	0.99004…	12.549
0.1	0.90483…	4.013

Wall-clock comparison (median of 5 trials, `mpmath` 80 dps; full table in `analysis/results/29_warnaar_17_ab_166`

β	L tri. (D)	ψ prod. (E)	Project C (F)	F vs D	F vs E	claim (vs D)	verdict
0.001	28.6 ms*	1,079 ms	0.0333 ms	**858**×*	**32,363**×	34×	CONFIRMED ($\geq 25\times$)
0.01	3.85 ms	10.4 ms	0.0334 ms	**118**×	**317**×	15×	CONFIRMED ($\geq 8\times$)
0.1	0.839 ms	1.89 ms	0.0309 ms	**26.7**×	**60**×	9×	CONFIRMED ($\geq 3\times$)

*Form D at $\beta = 10^{-3}$ returns -9.76×10^{96} instead of the true value $+39.638$—catastrophic cancellation across 10^{177} intermediate magnitudes at dps = 80. The 858× ratio measures the work the naive evaluator does, not the work to actually recover 30 digits. For 30-digit-correct recovery via form D, dps $\gtrsim 220$ is needed; form F (Project C) recovers 30 digits at dps = 80 in $\sim 33\,\mu$s.

Speedups. 27–858× vs the LHS triangular sum (caveat at $\beta = 10^{-3}$); 60–32,363× vs the product spelling near $q \to 1^-$. The claimed direction in [Hel26, diary 42] (9–34× vs LHS; "hundreds to tens of thousands" vs product) is confirmed; the measured *magnitude* of the LHS-comparison speedup exceeds the claim because form D at $\beta = 10^{-3}$ is much slower than the ChatGPT-Pro measurement reported (likely platform / version difference).

4.4. **A secondary point: numerical conditioning.** At $\beta = 10^{-3}$, the LHS hypergeometric summands reach magnitudes $\sim 10^{177}$ before cancelling to the true value ~ 40. Naive 30-digit computation of the LHS is *not just slow, it is numerically ill-conditioned*—large guard precision is required to recover the small cancelled value. The transformed form has no such issue: every term in the dual sum is bounded by 1.

This is an *additional* practical benefit of the transformation, on top of the wall-clock speedup: the engine is producing a form that is better-conditioned, not merely faster.

4.5. **Discussion.** This identity is more striking than Result 1 in one respect (the $\sim 10^{177}$ cancellation hazard) and less striking in another (the engine's basis cannot currently discover the Jacobi triple-product rewrite that takes $L(q)$ to the half-shifted Gaussian —that is a manual preprocessing step). Both shortcomings are addressed in v0.10 (see section 7.3).

5. Methodological note: the generalized obstruction principle

A central methodological pillar—closed at high confidence over three independent instances during the run [Hel26, diaries 19, 35, 37]—is the following:

Principle 5.1 (Generalized obstruction principle)**.** Scalar time-only functional calculus of a Fokker–Planck-type generator L preserves $\ker L$ (the equilibrium sector). Therefore, any Mellin-pair-style reciprocal-symmetric spectral object (Watson's $M(\alpha)$, Berndt–Dixit's $BD_{\text{int}}(\alpha)$, etc.) cannot be matched by a functional of L's equilibrium partition function under such a calculus.

The three confirming instances:

(1) **B1**—modular S-transform on a lattice-gas g^* vs Watson's $M(\alpha)$: closed via 20 candidate shadow functions across 5 structurally-different families all failing the strict $\leq 1\%$ / ≥ 2-dex bar [Hel26, diary 19]; cohomological irreducibility argument. The shadow g^* is an Eichler cocycle in $H^1(\Gamma, V_k)$.

(2) **B1.5**—Caputo time-fractional Fokker–Planck with harmonic confinement vs Berndt–Dixit $BD_{\text{int}}(\alpha)$: closed via a one-line argument (Mittag-Leffler $E_\sigma(0) = 1$ implies the equilibrium density is fixed pointwise under the subordination, so the partition function is σ-independent [Hel26, diary 35]).

(3) **B1.5-ext**—Riemann–Liouville time-fractional Fokker–Planck with *quartic* confinement vs Berndt–Dixit: closed via the same one-line argument with explicit handling of the RL initialization source term [Hel26, diary 37]. This sharpens the scope: the principle requires *scalar time-only* functional calculus. *Space-dependent* variable-order subdiffusion (Fedotov–Falconer literature) is the identified route to a potential counterexample.

5.1. **Implication for Project C.** The obstruction principle directly constrains Project C's search: *any operator sequence that attempts to bridge a partition-function- of-a-generator object on one side with a Mellin-pair-reciprocal spectral object on the other side is, generically, structurally ineligible.* The engine's basis does not include such bridge attempts (operators 1–3 all act *within* the analytic or modular sector, not across them); the principle gives a *justification* for the restriction, not just an empirical observation. A future `SectorClassifier` pre-pass (queued for v1.x) would make this restriction enforceable as a hard prune step (per section 6.1).

6. Connection to the toolkit-split cog-sci finding

The operator basis was distilled from empirical hand-tagging of Ramanujan's recorded work against a working vocabulary of operators. The current state (as of [Hel26, diary 36]; 200+ tagged proof units across four chapters of Berndt's commentaries and the Quarterly Reports) supports a *hierarchical-modular* decomposition (per [Hel26, §D1]):

Level	Cluster	Example chapters
Top	modular $\perp$ analytic	zero cross-firing in ~ 200 units
Sub (modular)	{q-shift, modular-S} for CFs	NB V Ch. 32
Sub (modular)	{modular-S} monoculture for modular eqs.	NB V Ch. 36
Sub (modular)	{class-field, modular-S} for class inv.	NB V Ch. 34
Sub (analytic)	Bipartite: Mellin/Master-Theorem $\perp$ Schröder–König	NB Part 1 Ch. 3–4, QRs

The bipartite analytic finding [Hel26, diary 36; BIG_IDEAS_REVIEW §2c] is that within the analytic toolkit, Mellin/Master-Theorem material and Schröder–König fractional iteration material fire in *disjoint entry-sets*— zero co-tagging across the 52 proof units that constitute the iteration and Master-Theorem subdomains of NB Part 1 Ch. 4.

6.1. **Implication for the engine.** The basis described in section 2 is the *engineering image* of this empirical decomposition: the three substantively-implemented operators correspond to the three highest-frequency operator clusters in the tagged corpus (modular family → operator 3; Mellin/Master-Theorem → operators 1–2; the four stub operators are placeholders for the Schröder–König sub-cluster, Frullani, Fourier self-reciprocity, and Mellin-pair reciprocity, which would be wired in v1.x).

The two results in sections 3 and 4 both use operator 3 (the modular family). This matches [Hel26, diary 36]'s finding that Lost Notebook entries in the partial-theta / theta-function neighbourhood (Warnaar's identity, $\psi(q)$) are exactly the targets a modular-cluster operator should hit.

A `SectorClassifier` pre-pass (queued v1.x, per [Hel26, diary 38]) would pin each new search state to either the modular or analytic sector up-front and constrain the proposer to operators from that sector only. By the toolkit-split hypothesis (zero cross-firing observed), this is expected to give a $\geq 3\times$ reduction in proposer-space search effort with no loss of coverage.

7. Discussion and future work

7.1. **What was demonstrated.** This paper reports *two specific Ramanujan identities where an automated symbolic-search engine, equipped with a small operator basis distilled from Ramanujan's empirically-observed working clusters, finds transformed computational forms that are* $13.7\times$ *to* $32{,}363\times$ *faster than the recorded forms at* q *near* 1^-. Both transformations use only the engine's existing `modular_transform_with_correction` operator (Result 1) and `mellin_inversion` operator (Result 2); both are verified to $\leq 6 \cdot 10^{-72}$ relative mismatch at 80-digit precision at the cross-check cell $\beta = 0.1$.

7.2. **What this is NOT.** It is important to be explicit:

(1) The identities are *recorded by Ramanujan* (Lost Notebook pp. 29 and 35) and have been *verified by Andrews, Berndt, Warnaar and others* in the standard commentaries. The engine did *not* discover them.
(2) The *transformed forms* are also classical in the sense that the underlying modular transformations (Dedekind-η, $\theta_2 \leftrightarrow \theta_4$) are textbook. The engine's contribution is to *compose them with the right preprocessing, verify numerically to high precision, and pass a cost gate that confirms the order-of-magnitude speedup is real.*
(3) The cost-ratio numbers in sections 3 and 4 have been *independently reproduced* in our `mpmath` environment [Hel26, diary 43] and are within a factor of $2\times$ of the originally claimed ChatGPT-Pro measurements [Hel26, diary 42] at every β cell.
(4) Both routes use *algebraic preprocessing* (parameter binding, Pochhammer rewrite for section 3; Jacobi triple-product rewrite for section 4) that is currently done by hand. Sections 4.3 and 7.3 discuss the v0.10 operator additions that would close this gap.

7.3. **Basis-completeness and the v0.10 roadmap.** As of v0.9, Project C reaches **8** of **12** seed targets at depth ≤ 4 under a budget of 10,000. The four unreachable targets—mock-theta f_3, partial-theta sums, finite Gauss-sum evaluations, and generic q-Pochhammer transforms—have no operator whose `apply_fn` currently returns non-`None` on them. v0.10 adds four operators:

(1) `mock_theta_completion_transform`: thin wrapper around the Zwegers $R_{a,b}$ erfc form [Hel26, diary 41]; see also [Zwe02]. Resolves a known methodological bug in our P2 prompt (which used a divergent integral form; the convergent Mordell form in `analysis/17_mock_theta_residual_ebm_v1.` is what our code actually uses).

(2) `partial_theta_completion_or_false_theta_transform`: completes partial-theta sums $\sum q^{n(n+1)/2}$ (optionally with $(-1)^n$).
(3) `gauss_sum_finite_evaluation`: cleanest of the four; quadratic Gauss-sum closed form with the silent-zero guard preserved (per the documented hazard at $q \equiv 2 \pmod 4$ in `rkc/registry.py`).
(4) `q_pochhammer_transform`: most general; lifts seed factors to recognised infinite-product form and applies Dedekind-η or Heine / q-binomial as appropriate. Would close the AB 1.6.6 preprocessing gap noted in section 4.3.

ChatGPT Pro estimates 1–2 days each [Hel26, diary 42]; with realistic test coverage and edge cases, more likely 3–7 days each. After v0.10: Project C would close the basis-completeness milestone for all 12 seed targets and unblock 4 registry families currently surfacing as `CompletionNotImplemented` in `rkc/completions.py`.

7.4. **The from-scratch-discovery milestone.** The natural next milestone—and one that this paper does *not* yet clear—is for the engine to produce an identity that is *not recorded anywhere in Ramanujan's notebooks or Berndt's commentaries*, and that holds to the same strict $\leq 10^{-30}$ acceptance bar across a $\geq$ 2-dex sweep. To our knowledge no automated identity-discovery engine has cleared this bar for q-series identities at the Ramanujan level of structure. Project C's roadmap to that milestone:

(a) Wire the v0.10 four operators above.
(b) Add the `SectorClassifier` pre-pass per section 6.1.
(c) Add a `CompressionFilter` per the metric in section 1 of this note (and [Hel26, diary 39]), with the grammar-relative caveats explicit.
(d) Run a directed sweep against seeds in the Lost Notebook neighbourhood (Warnaar 1.7 family, mock-theta f-family, etc.) at depth ≤ 6, budget $\sim 10^5$, with the engine's existing cost-gate set to require ratio $> 10\times$.

Whether (a)–(d) suffice for a from-scratch hit is the open question.

Acknowledgments

The eight ChatGPT-Pro Erdős-style prompt cycles [Hel26, diaries 35, 37, 38, 39, 40, 41, 42, plus the v0.8a proposer-test that preceded them] produced the structural insights and operator-completion proposals that drove the implementation reported here; their methodology (open-ended T/F prompts with anti-handwave guards and explicit numerical verification bars) is itself a piece of the methodology contribution. The standing-on-shoulders base is the immense body of work by B.C. Berndt, G.E. Andrews, S.O. Warnaar, F. Garvan, J. Frye, S. Zwegers, D. Hickerson, B.C. Berndt and A. Dixit, and the OEIS, without which none of the source identities, transformations, or verification points would exist. [Hel26, diary 41]'s detection of an error in our own P2 prompt (a divergent integral form picked up from training-memory rather than copied from our implementation) is a worked example of the "paste actual implementation, not memory-paraphrase" discipline that the ChatGPT-Pro methodology imposes.

Appendix A. The operator-3 apply-fn (excerpted from `rkc/search/basis.py`)

The pattern matchers for the two cases used in sections 3 and 4 are reproduced here in skeletal form (full implementation in `rkc/search/basis.py` at function `_modular_transform_with_correction_apply`):

```
def _modular_transform_with_correction_apply(expr, n):
    # CASE A: log Bose-Einstein  log(1 - exp(-beta n))
    #   -> log(1 - exp(-4 pi^2 n / beta))
    #      + correction (-pi^2/(6 beta) - (1/2) log(beta/(2 pi)) + beta/24)
    result = _log_bose_einstein_apply(expr, n)
```

```
    if result is not None:
        return result
    # CASE B: Lambert / E_2 quasi-modular  n q^n / (1 - q^n)
    #   -> -(4 pi^2 / beta^2) n / (exp(4 pi^2 n / beta) - 1)
    #      + correction pi^2/(6 beta^2) + 1/24 - 1/(2 beta)
    result = _lambert_e2_apply(expr, n)
    if result is not None:
        return result
    # CASE C: Stieltjes occupation-number (v0.8c)  1/(exp(beta n) - 1)
    result = _v08c_try_occupation_number_pattern(expr, n)
    if result is not None:
        return result
    return None
```

Each branch returns either `None` (declines) or a 3-tuple `(new_summand, additive_correction, summand_range)`. The engine populates the `OperatorApplication` from the tuple; the verifier picks up the correction and folds it into the before-side residual before comparison.

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